

SYEDA MASOOMA NAQVI

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RESEARCH INTERESTS

Human-Centered AI, Recovery from Substance Use Disorders, Online Communities, Human-Centered AI, ML Interpretability

EDUCATION

University of Minnesota, Twin Cities <i>Ph.D. Student in Computer Science (Human Centered AI)</i>	Aug 2024 – Present Minneapolis, MN
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University of Southern California <i>M.Sc. in Applied Data Science</i>	Sep 2022 – May 2024 LA, USA
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Lahore University of Management Sciences (LUMS) <i>B.S. in Economics and Mathematics (Joint)</i>	Aug 2017 – Jun 2021 Lahore, Pakistan
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PUBLICATIONS

Syeda Masooma Naqvi, Ruichen He, and Harmanpreet Kaur (2025). Catalyst for Creativity or a Hollow Trend?: A Cross-Level Perspective on The Role of Generative AI in Design. In Proceedings of the ACM Conference on Human Factors in Computing Systems (**CHI 2025**). Empirical study of how generative AI image tools reshape creativity, ownership, and design practice across experience levels.

Syeda Masooma Naqvi, Malik Khadar, Harmanpreet Kaur (**CHI 2025**) Computational Reification of Creativity: How Generative AI Reshapes Creative Cognition. Workshop on Tools for Thought at the ACM Conference on Human Factors in Computing Systems Theoretical framing of how generative AI alters creative cognition and the concept of authorship.

Syeda Masooma Naqvi, Gayatri Balaji, Angelique Laju and Harmanpreet Kaur. From Curiosity to Caution: How Expertise Shapes the Use of Interpretable Machine Learning (Accepted at **FAccT 2026**) Empirical evaluation of an interpretability tool with novice and expert data scientists; presents a proficiency-based framework with implications for responsible ML deployment.

TEACHING EXPERIENCE

University of Minnesota, Twin Cities Teaching Assistant · CSCI 4115, User Experience Design, Prototyping, and Evaluation · CSCI 4131, Internet Programming	Aug 2025 – Present
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University of Southern California · DSCI 510 Principles of Programming for Informatics	Aug 2023 – May 2024
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Lahore University of Management Sciences · SS 187 Applied Statistics · ECON-330 Econometrics · Econ-211 Intermediate Microeconomics	[Aug 2019 – May 2021]
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RESEARCH EXPERIENCE

Research Assistant

Dr. Lana Yarosh

Jan 2026 — Present

Recovery, Online Communities

- Conducting a large-scale quantitative study of Wikipedia’s page protection policy, analyzing editor behavior across 3,971 events using difference-in-differences methods.
- Participatory design workshops to co-design computational tools for youth in recovery. Collaboration with the longest running sober school in the country.

Research Assistant

Dr. Harmanpreet Kaur

June 2024 — Aug 2025

Interpretable ML, XAI

- Developed a proficiency-based framework characterizing expert-novice differences in Explainable AI; identified design implications for scaffolding appropriate reliance. (Forthcoming, FAccT 2026)
- Conducted artifact-based interviews (N=28) with designers at varying levels of generative AI tool exposure to study shifts in creativity, ownership, and professional workflow integration. (CHI 2025)
- Synthesized qualitative findings into structured design recommendations for responsible, human-centered AI systems.

Research Assistant

Dr. Maryam Mustafa, Dr. Agha Ali Raza

Oct 2022 — May 2023

ICT4D, Social Computing

- Designed and deployed a character-based storytelling intervention across public WhatsApp groups in Pakistan to measure the effectiveness of public service messaging for low-literate, hard-to-reach populations. Used baseline and endline surveys to quantify shifts in user knowledge and awareness.
- Used a pre-post study on an IVR platform in Pakistan to measure belief updating across socio-cultural topics using MCQs administered to users assigned to one of three intervention modalities: quiz, short message, or story.

Research Assistant

Center for Economic Research Pakistan

Jan 2021 – May 2022

Economics of Education

- Evaluated the effect of positive psychology exercises on academic outcomes using pre/post performance metrics, contributing to evidence on scalable, low-cost school interventions.

SKILLS

Programming	Python, R, Stata, SQL, C++, Javascript
Packages	PyTorch, HuggingFace Transformers, scikit-learn
Tools	Git, Slurm, L ^A T _E X, AWS, VSCode & Copilot
Visualization	Tableau, Power BI, matplotlib, ggplot
Methods	Interviews, Surveys, Contextual Inquiry, RCTs, Log Analysis, Causal Inference, Statistical Analysis, Econometrics, Machine Learning

FELLOWSHIPS AND HONORS

Fulbright Scholarship 2022-2024

Graduation with High Merit 2021

Dean’s Honour List, LUMS

Afiniti Limited Scholarship 2017-2018